

Q1: Continuous Improvement Actions based on SO Assessment

Student outcomes evaluated by the direct assessment method provide input for developing recommendations and actions for continuous improvement. From the course rubric assessments, the instructors analyze the average scores and the number of students attaining a satisfactory, developing or unsatisfactory ratings. Based on these results, recommendations for continuous improvements are made and incorporated into the courses. Samples of course rubric assessment, results, and continuous improvement actions are given in the Q1: Documentation Folder. The Senior Exit survey (assessment method #4) is also used as an indirect method to determine how well the students are perceiving that they are attaining the SOs. The ratings from the senior exit survey correlate well with the direct assessment methods.

The Engineering Expo judging rubrics are also used to assess the attainment of the SO's by the students on a yearly basis. Based on the analysis from the rubrics on the capstone projects in Spring 2020, the CHE 397 senior design instructor, Professor Bilgin, recommended several continuous improvement actions that will be incorporated into CHE 397 senior design course to improve the capstone project team's project. (see supporting file list below)

Based on our Level 3 and Level 4 analysis of the attainment by different assessment methods and attainment for all SO's, the department initiated a 'Curriculum Day' to discuss which aspects of the curriculum need of improvement and to recommend action items. The Curriculum Day was held in Fall 2018 with a majority of faculty present. The meeting consisted of faculty discussing the courses they teach and how to improve the content and student attainment of the SOs, with feedback from the rest of the faculty. In addition, there were 5 topic-specific lectures; 'Best Practices in Teaching' by Professor Wedgewood, 'Best Practices in mentoring UGs for research' by Professor Sharma, 'Teaching Methods' by Professor Bilgin, a lecture on one of our extracurricular student learning experiences, 'Lessons from the Chem-E-Car Experience', by Professor Singh and a Discussion on Improving student experience, led by Professors Berry and Nitsche. The meeting resulted in several action items to improve the curriculum. The Curriculum Day agenda and minutes with action items are given in the Q1: Documentation folder. The department plans on holding a Curriculum Day every few years to improve our program based on our Level 3/Level 4 SO assessment.

List of Supporting Files in Q1: Documentation Folder:

course_assessment_che201F17_zdunek_assignment_rubric_results.pdf

course_assessment_che205Fa19_mehraeen_assignment_rubric_results.pdf

course_assessment_che210Fall19_wedgewood_assignment_rubric_results.pdf

course_assessment_che321F19_singh_assignment_rubric_results.pdf

course_assessment_che381F15_zdunek_assignment_rubric_results.pdf

course_assessment_che201F17_zdunek_assignment_rubric_results.pdf

expo_rubric_calculations_2020_actions.pdf

faculty_curriculum_day_agenda_2018.pdf

faculty_curriculum_day_minutes_2018.pdf